PRS Pricing Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 22-March-2026

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1 Test Results — PRS Pricing Model (MKM-PR-001)

Tests run on **22 March 2026 at 17:02:41**.

Table 1: Test Results Summary — PRS Pricing Model (MKM-PR-001)

Total Tests	61
Passed	61
Failed	0
Skipped	0
Duration	9.78s

Table 2: Detailed Test Results — PRS Pricing Model

Test Description	Acceptance Criteria	Result	Time
Basis waterfall in generated output	`bw1["model_uncertainty_bp"] == 2.0; bw1["terrain_bp"] == 2.0` (+6 more)	Pass	0.554s
Composition flat penalty	`result["composition_bp"] == COMPOSITION_BASIS_BPS["Flat"]`	Pass	0.000s
Composition pre2000 penalty	`result["composition_bp"] == 1.0`	Pass	0.000s
Distance basis capped	`result["distance_bp"] == DISTANCE_MAX_BPS`	Pass	0.000s
Distance basis scales linearly	`result["distance_bp"] == pytest.approx(1.0, abs=0.1)`	Pass	0.000s
Elevation above gauge reduces spread	`result["elevation_bp"] == pytest.approx(-1.0, abs=0.1)`	Pass	0.000s
Elevation below gauge no benefit	`result["elevation_bp"] == 0.0`	Pass	0.000s
Elevation capped at 3bp	`result["elevation_bp"] == -ELEVATION_MAX_BENEFIT_BPS`	Pass	0.000s
Model uncertainty is fixed	`result["model_uncertainty_bp"] == MODEL_UNCERTAINTY_BPS`	Pass	0.000s
Terrain zone1 low risk	`result["terrain_bp"] == 0.0`	Pass	0.000s
Terrain zone3 high risk	`result["terrain_bp"] == TERRAIN_BASIS_BPS["Zone 3a"]`	Pass	0.000s
Total basis sums components	`result["total_basis_bp"] == pytest.approx(expected_total,...`	Pass	0.000s
1Y from Feb 2026 = last Friday May 2027.	`mat == last_friday_of_month(2027, 5)`	Pass	0.000s
3Y from Feb 2026 = last Friday May 2029.	`mat == last_friday_of_month(2029, 5); mat.month == 5` (+1 more)	Pass	0.000s
Jan 2027 uses Aug 2026 roll → Nov 2029 maturity.	`mat == last_friday_of_month(2029, 11)`	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
3Y from Sep 2026 = last Friday Nov 2029 (Aug roll).	`mat == last_friday_of_month(2029, 11)`; `mat.month == 11`	Pass	0.000s
5Y from Feb 2026 = last Friday May 2031.	`mat == last_friday_of_month(2031, 5)`	Pass	0.000s
5Y from Sep 2026 = last Friday Nov 2031.	`mat == last_friday_of_month(2031, 11)`	Pass	0.000s
All maturities should fall on a Friday.	`mat.weekday() == 4`	Pass	0.000s
3Y maturity jumps from May to Nov at the Aug roll.	`may_mat.month == 5`; `nov_mat.month == 11` (+1 more)	Pass	0.000s
Aug onwards uses Aug roll.	`roll == last_friday_of_month(2026, 8)`; `roll.month == 8`	Pass	0.000s
December uses the same year's Aug roll.	`roll.month == 8`; `roll.year == 2026`	Pass	0.000s
Feb-Jul uses Feb roll.	`roll == last_friday_of_month(2026, 2)`; `roll.month == 2`	Pass	0.000s
January uses the previous year's Aug roll.	`roll == last_friday_of_month(2026, 8)`; `roll.month == 8` (+1 more)	Pass	0.000s
July is the last month of the Feb roll period.	`roll.month == 2`; `roll.year == 2026`	Pass	0.000s
March is still in the Feb roll period.	`roll.month == 2`; `roll.year == 2026`	Pass	0.000s
compute_maturity_date with no ref_date uses today.	`isinstance(mat, date)`; `mat.weekday() == 4`	Pass	0.000s
current_roll_date(None) uses today and returns a valid date.	`isinstance(roll, date)`; `roll.weekday() == 4`	Pass	0.000s
maturity_table with no ref_date uses today.	`len(table) == 5`; `all(isinstance(e['maturity_date'], date) for e in table)`	Pass	0.000s
Empty propertyts dir	`stats["total_properties"] == 0`; `stats["properties_with_gev"] == 0`	Pass	0.002s
Generator should still work without gauge hazard curves (empty basis).	`stats["properties_with_gev"] == 2`; `stats["properties_with_floor"] == 1` (+1 more)	Pass	0.583s
No propertyts dir	—	Pass	0.000s
Aug 2026: 31st is Monday, last Friday = 28th.	`last_friday_of_month(2026, 8) == date(2026, 8, 28)`	Pass	0.000s
Feb 2026: 28th is Saturday, last Friday = 27th.	`last_friday_of_month(2026, 2) == date(2026, 2, 27)`	Pass	0.000s
May 2029: 31st is Thursday, last Friday = 25th.	`last_friday_of_month(2029, 5) == date(2029, 5, 25)`	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Nov 2029: 30th is Friday.	`last_friday_of_month(2029, 11) == date(2029, 11, 30)`	Pass	0.000s
Result should always be a Friday (weekday=4).	`d.weekday() == 4`	Pass	0.000s
Actual years should be strictly increasing.	`years[i] > years[i - 1]`	Pass	0.000s
All maturities should be in May for the Feb roll.	`e['maturity_date'].month == 5`	Pass	0.000s
All maturities should be in November for the Aug roll.	`e['maturity_date'].month == 11`	Pass	0.000s
Maturity string should be formatted as 'DD Mon YYYY'.	`len(e['maturity_str']) > 8; 'May' in e['maturity_str'] or 'Nov' in e['maturity_str']`	Pass	0.000s
Default table has 5 entries (1Y through 5Y).	`len(table) == 5`	Pass	0.000s
Tenors should be 1, 2, 3, 4, 5.	`tenors == [1, 2, 3, 4, 5]`	Pass	0.000s
Positive spread	`spread > 0; 50 < spread < 500`	Pass	0.000s
Spread increases with hazard	`spread_high > spread_low`	Pass	0.000s
Spread values at different tenors	`s > 0`	Pass	0.000s
Zero hazard rate	`spread == 0.0`	Pass	0.001s
Basis calculation	`len(nearest) > 0; ng0["gauge_id"] == "GAUGE-001"` (+5 more)	Pass	0.797s
Depth thresholds	`"any_flood" in thresholds; "moderate" in thresholds` (+4 more)	Pass	0.566s
Generate produces output	`output_path.exists(); "metadata" in data` (+2 more)	Pass	0.594s
Gev params structure	`"shape" in gev; "loc" in gev` (+2 more)	Pass	0.601s
High transmission rate	`prop3["summary"]["flood_transmission_rate"] == 1.0`	Pass	1.027s
Output structure	`"PROP-001" in curves; "PROP-002" in curves` (+21 more)	Pass	0.568s
Properties with gev	`stats["properties_with_gev"] == 2; stats["properties_with_floor"] == 1` (+1 more)	Pass	0.573s
Survival decreases with tenor	`survival[i] <= survival[i - 1]`	Pass	2.482s
Term structure	`ts["tenors"] == TENORS; threshold_name in ts` (+3 more)	Pass	0.540s
Recovery rates by trigger	`len(spreads) > 0`	Pass	0.892s
Recovery reduces spread	`spread_with_recovery < spread_no_recovery`	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All tenors should round-trip through maturity and back.	`tenor_from_maturity(mat, ref) == t`	Pass	0.000s
A date not in the table returns the closest tenor.	`result in [1, 2, 3, 4, 5]`	Pass	0.000s
Exact maturity date returns correct tenor.	`tenor_from_maturity(mat, date(2026, 2, 24)) == 3`	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (30 tests)	D. Kelly
07-Mar-2026	1.1	57 test(s) added; 30 test(s) removed (total: 57)	D. Kelly
07-Mar-2026	1.2	30 test(s) added (total: 87)	D. Kelly
18-Mar-2026	1.3	4 test(s) added (total: 91)	D. Kelly
22-Mar-2026	1.4	30 test(s) removed (total: 61)	D. Kelly